

 Adjustment for operation with light load

8. Start the motor without load / with light load and operate to 1/2 of the rated rotor speed
 - A1. Start operation direction:
 - a. If the start operation direction is wrong
SPM: increase the current proportion for Pr.10-42 (initial angle detection pulse value) to improve the accuracy of the angle detection.
IPM: Increase the voltage for Pr.10-52 (injection magnitude) to improve the accuracy of the angle detection.
 - b. If an ocA error occurs when pressing RUN to start the motor, decrease the current proportion for Pr.10-42 (initial angle detection pulse value).
 - A2. Operates the motor in 1/2 of the rated rotor speed, adjust the no-load operating current
If the no-load operating current exceeds 20% of the rated current, increase Pr.07-26 (torque compensation gain) and observe the no-load operating current.
 - A3. Accelerate to the rated frequency and observe if the motor operates stably.
 - a. If the motor output rotor speed presents periodic low-frequency wave, increase Pr.10-34 (PM sensorless speed estimator low-pass filter gain), or increase Pr.10-32 (PM FOC sensorless speed estimator bandwidth).
 - b. If the output frequency reflects high frequency vibration, decrease Pr.10-34 or decrease Pr.10-32.
 - A4. Accelerate the motor to the maximum rotor speed and observe if it operates stably.
If the motor stalls when accelerating to the maximum rotor speed, then increase Pr.10-34 (PM sensorless speed estimator low-pass filter gain), or increase Pr.00-17 (carrier frequency, you must set the carrier frequency larger than 10 times of the maximum output frequency)